ENVS 193DS Final

Owen Wright

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GitHub repository: https://github.com/owenwright-oss/ENVS-193DS_spring-2026_final

Set up

```
#Load packages
library(tidyverse)
library(here)
library(janitor)
library(ggeffects)
library(DHARMa)
library(gt)

#Read in data
nests <- read_csv(here("data", "nest_data_final.csv"),
                        show_col_types = FALSE)

#Clean names
nests <- clean_names(nests)

#Read in personal data
my_data <- read_csv(here("data",
                         "ENVS193DS_Homework2_personal_data - Sheet1
                          ↪  (1).csv"),
                    show_col_types = FALSE)

#Clean names of personal data
my_data <- clean_names(my_data)
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a binomial generalized linear model, or a logistic regression, to test whether distance to water edge predicts the probability of American Avocet microhabitat use. In part 2, they used a chi-square test of independence in order to test if the two categorical variables, bird species and habitat type, were related.

b. Suggestions for rewriting

American Avocet microhabitat use was found to be associated with distance to the water edge, meaning that American Avocets were more likely to use certain areas depending on how close those areas were to the waters edge in the San Francisco Bay. Bird species and habitat type was also found to be associated, suggesting that American Avocets, Forster's Terns, and Black-necked Stilts were not evenly distributed across managed ponds, non-tidal marshes, salt ponds, tidal marshes, and tidal mudflats habitats.

c. Further analysis

One additional variable that may influence the probability of American Avocet microhabitat use is total vegetation cover, measured as a percent of the ground covered by vegetation within a fixed radius. The data may be collected via aerial imagery around each bird observation point. Percent vegetation cover is a continuous variable, and may influence American Avocet habitat use as variation in vegetation density may affect predator and feeding behaviors by altering visibility or access to part of the habitat.

Problem 2. Data analysis

a. Clean and wrangle

```
#Create clean nest data
nests_clean <- nests |>

  #Select columns
  select(case_control,
         height_cm,
         ant_species) |>

  #Filter to the three ant species
  filter(ant_species %in% c("AB",
                            "RRB",
                            "BBR")) |>

  #Reorder levels of ant species
  mutate(ant_species = factor(ant_species,
                              levels = c("AB",
                                         "RRB",
```

```
                                          "BBR")),

         #Create column for full scientific names
         ant_species_full = case_when(ant_species == "AB" ~ "Crematogaster
          ↪  sjostedti",
                                      ant_species == "RRB" ~ "Crematogaster
 ↪  mimosae",
                                      ant_species == "BBR" ~ "Crematogaster
 ↪  nigriceps"),

         #Reorder full scientific name levels
         ant_species_full = factor(ant_species_full,
                                   levels = c("Crematogaster sjostedti",
                                              "Crematogaster mimosae",
                                              "Crematogaster nigriceps")))

#Show structure of cleaned data
glimpse(nests_clean)
```

```
Rows: 270
Columns: 4
$ case_control     <dbl> 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1,~
$ height_cm        <dbl> 391.5, 310.0, 513.0, 154.0, 324.5, 413.0, 167.0, 276.~
$ ant_species      <fct> RRB, RRB, RRB, RRB, AB, RRB, BBR, RRB, AB, BBR, RRB, ~
$ ant_species_full <fct> Crematogaster mimosae, Crematogaster mimosae, Cremato~
```

```
#Show 10 random rows from data frame
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
   case_control height_cm ant_species ant_species_full
          <dbl>     <dbl> <fct>       <fct>
 1            0      130  RRB         Crematogaster mimosae
 2            0      316  RRB         Crematogaster mimosae
 3            0      169  BBR         Crematogaster nigriceps
 4            0      134  RRB         Crematogaster mimosae
 5            0      242. RRB         Crematogaster mimosae
 6            0      131  RRB         Crematogaster mimosae
 7            1      413  RRB         Crematogaster mimosae
 8            1      201  RRB         Crematogaster mimosae
 9            1      238. RRB         Crematogaster mimosae
```

```
10              0      298. RRB         Crematogaster mimosae
```

b. Response variable

In this data set, biologically, a value of 1 means that a tree contained a bird nest and was therefore selected by a bird as a suitable nest habitat. A value of 0 means that the tree was classified as an available nesting tree but did not contain a bird nest, and was therefore not selected by a bird to serve as a nest site despite this.

c. Purpose of study

The first predictor variable, tree height, is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence may increase as taller trees may provide more suitable nesting structure and protection from predators. This information was found in the first paragraph of the "Methods" section, and further articulated in the first paragraph of the "Results" section, where it is reported that all nests were in trees more than 1.5 meters tall.

The second predictor variable, acacia ant species, is a categorical variable, represented by three main species: *Crematogaster sjostedti*, *Crematogaster mimosae*, and *Crematogaster nigriceps*. Ant species may influence the probability of nest occurrence as some species, such as *C. mimosae* and *C. nigriceps*, aggressively defend their host trees, which may make those trees a more secure nesting site. This information was found in the second paragraph of the "Introduction" section, and further articulated in the first paragraph of the "Discussion" section, where it is explained that birds almost always nested in trees inhabited by aggressive ant defenders.

d. Table of models

```
#Create table of models
model_table <- tibble(model_number = c("Model 1",
                                       "Model 2"),
                      response_variable = c("Nest occurrence",
                                            "Nest occurrence"),
                      predictor_variables = c("Tree height + acacia ant
                       ↪  species",
                                              "Tree height + acacia ant
                                               ↪  species"),
                      interaction_term = c("No interaction",
                                           "Height × ant species"))
```

Model number	Response variable	Predictor variables	Interaction term
Model 1	Nest occurrence	Tree height + acacia ant species	No interaction
Model 2	Nest occurrence	Tree height + acacia ant species	Height × ant species

```
#Display table of models
model_table |>
  gt() |>
  cols_label(model_number = "Model number",
             response_variable = "Response variable",
             predictor_variables = "Predictor variables",
             interaction_term = "Interaction term")
```

e. Run the models

```
#Do not display any output
#| output: false

#Run model without interaction
mod_no_int <- glm(case_control ~ height_cm + ant_species_full,
                  data = nests_clean,
                  family = "binomial")

#Run model with interaction
mod_int <- glm(case_control ~ height_cm * ant_species_full,
               data = nests_clean,
               family = "binomial")
```

f. Select the best model

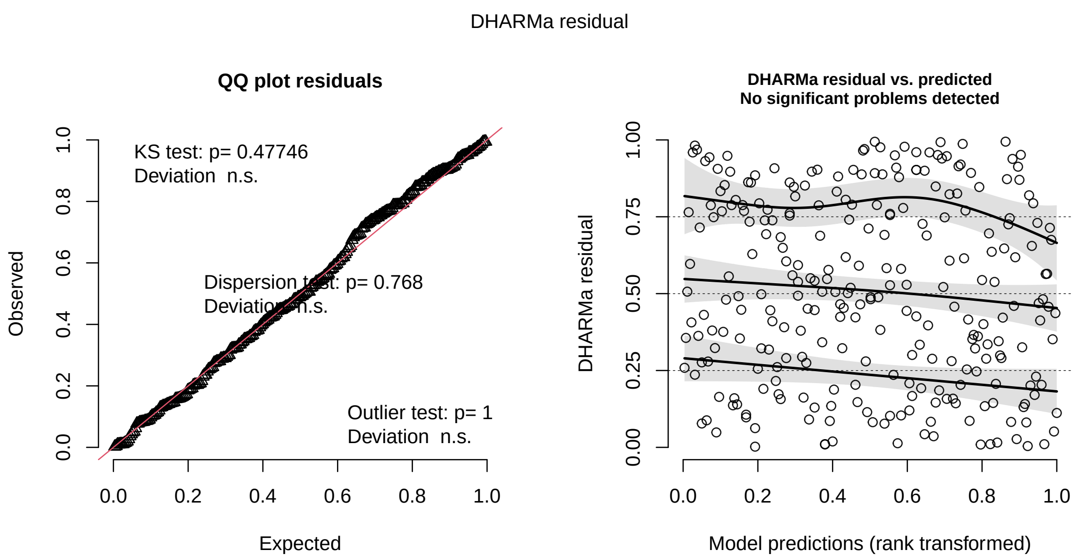
```
#Compare models using Akaike's Information Criterion
aic_table <- AIC(mod_no_int,
                 mod_int)

#Display AIC table
aic_table
```

```
            df      AIC
mod_no_int   4 258.7430
mod_int      6 237.8042
```

The best model, as determined by Akaike's Information Criterion (AIC), was found to be the logistic regression model predicting nest occurrence based on tree height and acacia ant species, as well as the interaction between tree height and acacia ant species. This model had the lower AIC value, suggesting that the effect of tree height on probability of nest occurrence differs depending on which acacia ant species is present.

g. Check the diagnostics

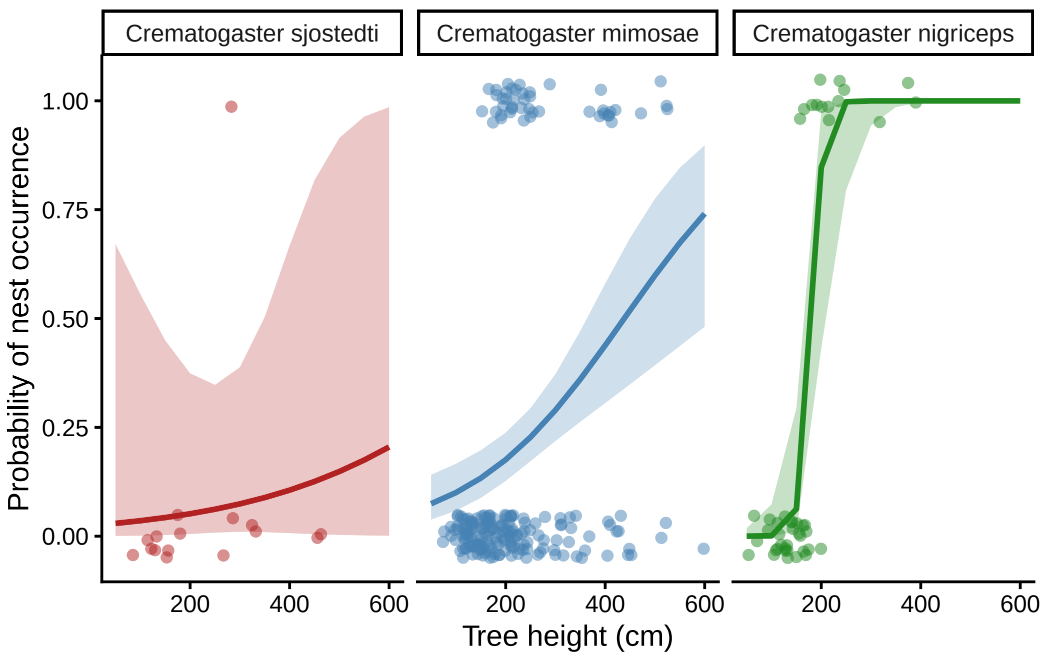
```r
#Set seed to keep same diagnostic plot
set.seed(123)

#Create simulated residuals for model 2
mod_int_res <- simulateResiduals(fittedModel = mod_int)

#Display DHARMa diagnostic plots
par(mar = c(5, 5, 4, 2),
    cex = 0.75)
plot(mod_int_res)
```



h. Visualize the model predictions

```
mod_int_preds <- suppressMessages(
  ggpredict(mod_int,
            terms = c("height_cm",
                      "ant_species_full")))

#Convert predictions to data frame
mod_int_preds <- as.data.frame(mod_int_preds)

#Rename group column to match raw data
mod_int_preds <- mod_int_preds |>
  rename(ant_species_full = group)

#Plot model predictions and raw data
ggplot() +

  #Add underlying data
  geom_jitter(data = nests_clean,
              aes(x = height_cm,
                  y = case_control,
                  color = ant_species_full),
              width = 0,
              height = 0.05,
              alpha = 0.5) +

  #Add confidence intervals around predictions and underlying data
  geom_ribbon(data = mod_int_preds,
              aes(x = x,
                  ymin = conf.low,
                  ymax = conf.high,
                  fill = ant_species_full),
              alpha = 0.25) +

  #Add model prediction lines
  geom_line(data = mod_int_preds,
            aes(x = x,
                y = predicted,
                color = ant_species_full),
            linewidth = 1) +

  #Create separate panels for each ant species
```

```r
facet_wrap(~ ant_species_full) +

#Change colors
scale_color_manual(values = c("firebrick",
                              "steelblue",
                              "forestgreen")) +
scale_fill_manual(values = c("firebrick",
                             "steelblue",
                             "forestgreen")) +

#Label axes in full
labs(x = "Tree height (cm)",
     y = "Probability of nest occurrence") +

#Remove legend
theme_classic() +
theme(legend.position = "none")
```



i. Write a caption for your figure

Figure 1. Predicted probability of nest occurrence by tree height and acacia ant species. Points represent the underlying nest occurrence data, where 1 indicates a tree

containing a bird nest and 0 indicates an available and suitable tree that does not contain a bird nest. The lines represent model predictions from the selected logistic regression model, and surrounding lighter coloured shaded ribbons represent the 95% confidence intervals. Separate panels show predictions for *Crematogaster sjostedti* (displayed in red), *Crematogaster mimosae* (displayed in blue), and *Crematogaster nigriceps* (displayed in green). Data are from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna https://doi.org/10.5281/zenodo.8373322

j. Calculate model predictions

```
#Create new data for a tree height fixed at 600 cm
pred_600 <- tibble(height_cm = c(600,
                                 600,
                                 600),
                   ant_species_full = factor(c("Crematogaster sjostedti",
                                               "Crematogaster mimosae",
                                               "Crematogaster nigriceps"),
                                             levels = c("Crematogaster
                                              ↪  sjostedti",
                                                        "Crematogaster
                                                         ↪  mimosae",
                                                        "Crematogaster
                                                         ↪  nigriceps")))

#Calculate model predictions at 600 cm
pred_600_output <- predict(mod_int,
                           newdata = pred_600,
                           type = "link",
                           se.fit = TRUE)

#Create table by converting predictions from log-odds into probabilities
pred_600_table <- pred_600 |>
  mutate(predicted_probability = plogis(pred_600_output$fit),
         conf_low = plogis(pred_600_output$fit - 1.96 *
          ↪  pred_600_output$se.fit),
         conf_high = plogis(pred_600_output$fit + 1.96 *
          ↪  pred_600_output$se.fit))

#Display the predicted probabilities with all columns
pred_600_table |>
```

```
#Show columns in order
select(ant_species_full,
       height_cm,
       predicted_probability,
       conf_low,
       conf_high)
```

```
# A tibble: 3 x 5
  ant_species_full        height_cm predicted_probability conf_low conf_high
  <fct>                       <dbl>                 <dbl>    <dbl>     <dbl>
1 Crematogaster sjostedti       600                 0.205 0.000970     0.986
2 Crematogaster mimosae         600                 0.741 0.481        0.898
3 Crematogaster nigriceps       600                 1     1.000        1
```

k. Interpret your results

When tree height is set to 600 cm, the model predicted that the probability of nest occurrence was highest for trees inhabited by the more aggressive *Crematogaster nigriceps* at 1.000 (95% CI: 1.000, 1.000), followed by the *Crematogaster mimosae* at 0.741 (95% CI: 0.481, 0.898), and was the lowest for the *Crematogaster sjostedti* at 0.205 (95% CI: 0.001, 0.986). As displayed in Figure 1, the relationship between tree height and probability of nest occurrence differed depending on the type of acacia ant species present, with nest occurrence increasing most strongly with height for trees inhabited by the *C. nigriceps* ants. Ultimately, nest occurrence was more likely in that of taller trees and in trees inhabited by the more aggressive ant defenders, such as *C. nigriceps* and *C. mimosae*. Biologically speaking, this pattern may exist as aggressive ant species better defend host trees from predators and disturbances, making these trees better suited for nest selection.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

The affective visualization was primarily inspired by the scatter plot exploratory visualization examining the relationship between sleep duration and productivity level. The heights of the buildings represented sleep duration, while exposure to sunlight represented productivity. The arrangement of the buildings follows the overall trendline displayed in the scatter plot, allowing the affective visualization to still effectively communicate the relationship between the two variables but in a more visually compelling manner.

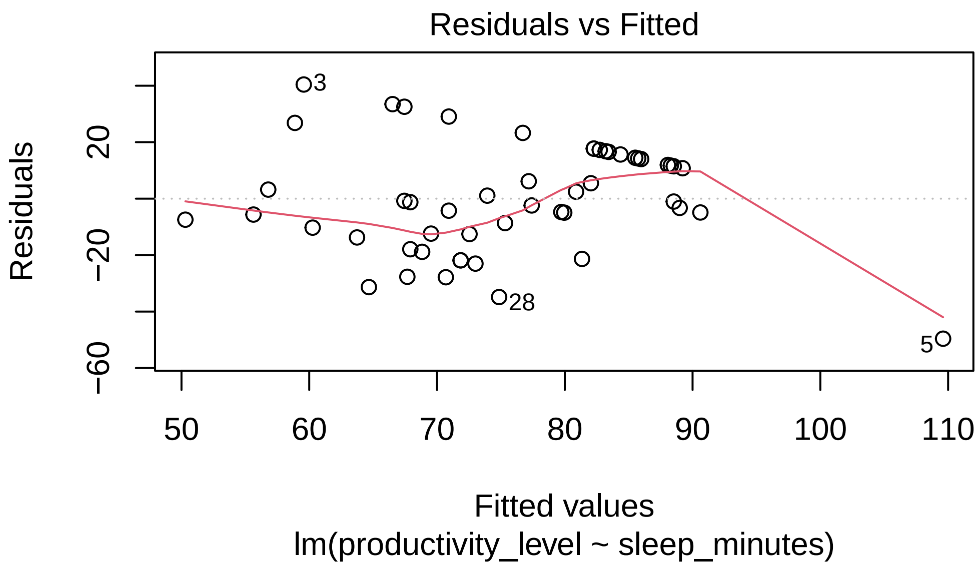
b. Designing an analysis

A simple linear regression model may be used to answer my initially proposed research question, "How does the amount of sleep I get a night affect my daily productivity?" This analysis is appropriate as the measured response variable, productivity level (measured as tasks completed over tasks initially planned and expressed as a percentage), and the predictor variable, sleep duration (measured in minutes), are both continuous variables. This analysis would ultimately allow me to determine if changes in my sleep duration are associated with changes in my overall productivity level, and further articulate the strength and direction of that relationship.

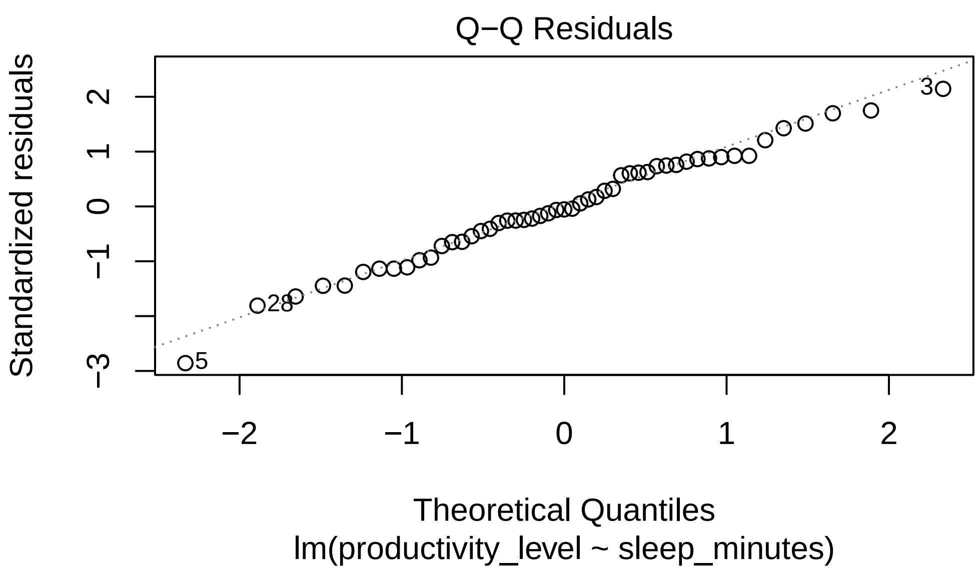
c. Check any assumptions and run your analysis

Check any assumptions

```
#Create productivity variable as percentage
my_data_clean <- my_data |>
  mutate(productivity_level = tasks_completed / tasks_planned * 100)

#Create linear model
sleep_productivity_mod <- lm(productivity_level ~ sleep_minutes,
                             data = my_data_clean)

#Check linearity and if variance is constant
plot(sleep_productivity_mod, which = 1)
```




```r
#Check if residuals are normal
plot(sleep_productivity_mod, which = 2)
```



Run the analysis proposed in part b

```
#Run simple the linear regression
summary(sleep_productivity_mod)
```

```
Call:
lm(formula = productivity_level ~ sleep_minutes, data = my_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max 
-49.608 -12.467  -1.028  14.368  40.428 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)   -24.28496   24.85887  -0.977 0.333411    
sleep_minutes   0.23165    0.05724   4.047 0.000184 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.46 on 49 degrees of freedom
Multiple R-squared:  0.2505,	Adjusted R-squared:  0.2352 
F-statistic: 16.38 on 1 and 49 DF,  p-value: 0.0001836
```

Calculate an effect size

```
#Calculate effect size for linear regression
summary(sleep_productivity_mod)$r.squared
```

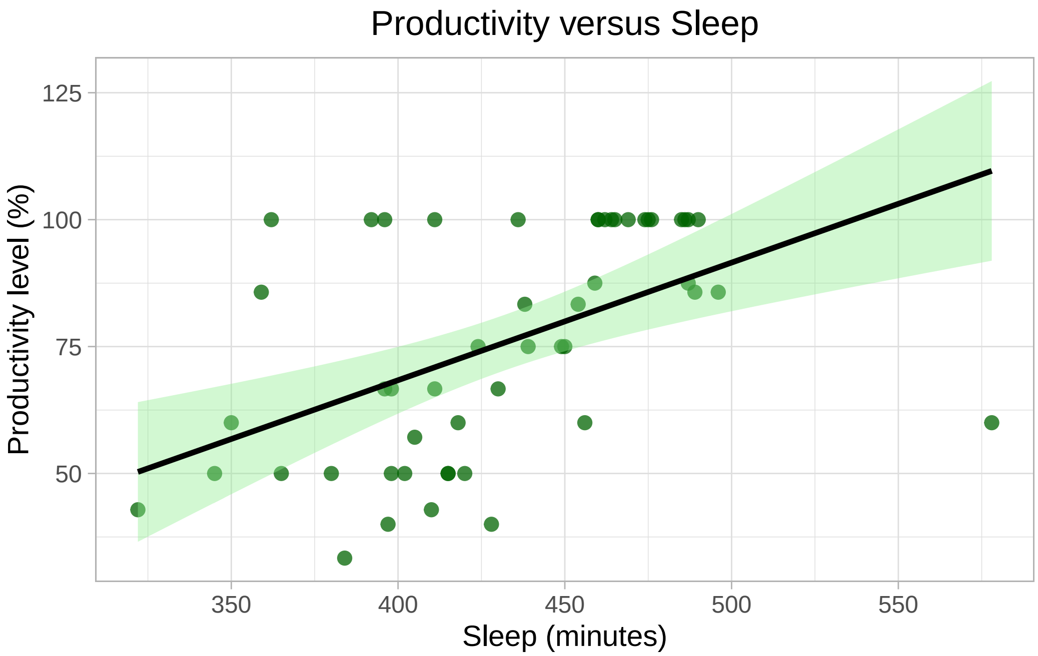
```
[1] 0.2505394
```

d. Create a visualization

```
#Create a scatterplot to display relationship between sleep and productivity
ggplot(data = my_data_clean,
       mapping = aes(x = sleep_minutes,
```

```
                    y = productivity_level)) +

  #Add one point for each daily observation from data set
  geom_point(color = "darkgreen",
             size = 2,
             alpha = 0.75) +

  #Add linear regression trend line from analysis and 95% confidence interval
  geom_smooth(method = "lm",
              se = TRUE,
              color = "black",
              fill = "lightgreen") +

  #Relabel both axes
  labs(x = "Sleep (minutes)",
       y = "Productivity level (%)",
       title = "Productivity versus Sleep") +

  #Set x-axis tick marks
  scale_x_continuous(breaks = seq(300,
                                  600,
                                  by = 50)) +
  #Change theme
  theme_light() +

  #Center title
  theme(plot.title = element_text(hjust = 0.5))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



e. Write a caption

Figure 2. Relationship between sleep duration and productivity level. Green points represent daily observations, derived from my personal data set, where sleep duration is measured in minutes slept each night and productivity level is measured as tasks completed divided by tasks planned, expressed as a percentage. The black line displays the predicted relationship from the simple linear regression model, and the surrounding light green shaded ribbon represents the 95% confidence interval. Data are from my personal data collected in ENVS 193DS at UCSB during Spring 2026.

f. Write about your results

A simple linear regression model was used to analyze the relationship between sleep duration and productivity level. The relationship was found to be statistically significant (simple linear regression, $F(1, 49)=16.38$, $p<0.001$, $\alpha = 0.05$, $R^2=0.251$), with sleep duration explaining around 25.1% of the variation seen in productivity level. As displayed in Figure 2, an increase in productivity level coincided with an increase in sleep duration, with each additional minute of sleep predicting an increase of 0.232 percentage points in productivity level.